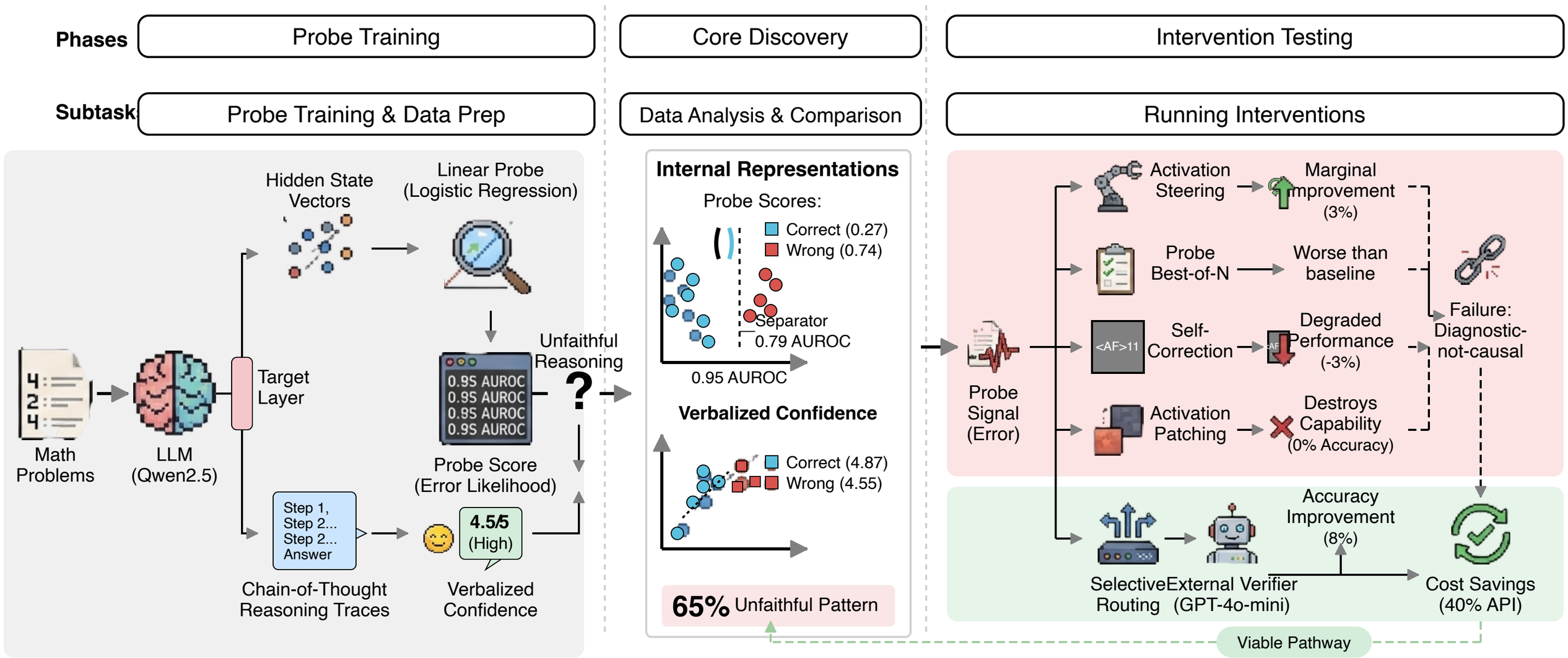

Phases
Probe Training
Core Discovery
Intervention Testing
Subtask
Probe Training & Data Prep
Data Analysis & Comparison
Running Interventions
Hidden State Vectors
Linear Probe (Logistic Regression)
Math Problems
LLM (Qwen2.5)
Target Layer
Unfaithful Reasoning
0.95 AUROC
0.95 AUROC
0.95 AUROC
0.95 AUROC
Probe Score (Error Likelihood)
Step 1,
Step 2...
Step 2...
Answer
4.5/5 (High)
Chain-of-Thought Reasoning Traces
Verbalized Confidence
Internal Representations
Probe Scores:
Correct (0.27)
Wrong (0.74)
Separator
0.79 AUROC
0.95 AUROC
Verbalized Confidence
Correct (4.87)
Wrong (4.55)
65% Unfaithful Pattern
Probe Signal (Error)
Activation Steering
Marginal Improvement (3%)
Probe Best-of-N
Worse than baseline
<AF>11
Self-Correction
Degraded Performance (-3%)
Activation Patching
Destroys Capability (0% Accuracy)
Failure: Diagnostic-not-causal
Selective Routing
External Verifier (GPT-4o-mini)
Accuracy Improvement (8%)
Cost Savings (40% API)
Viable Pathway